

Finding the students who will leave, while there is still time to help

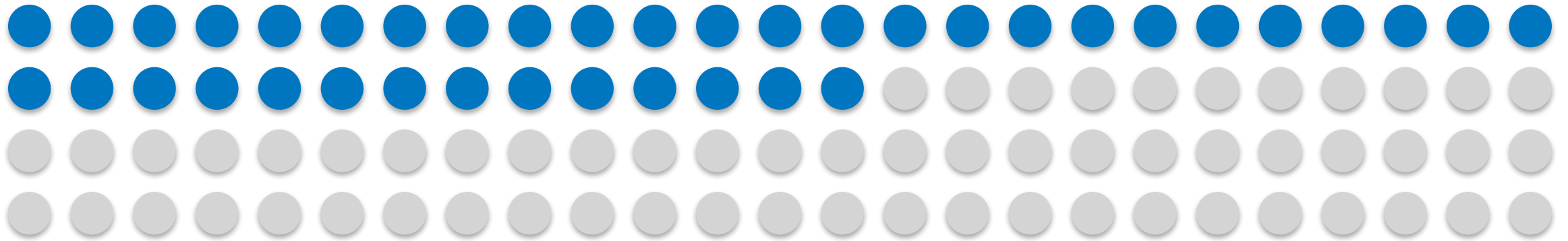
An early warning on day one. And one decision only the school can make.

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PGD AI/ML capstone, Pillar 5

Built on 4,424 student records from Instituto Politecnico de Portalegre

391 students in every 1,000 leave before they finish



One dot is ten students. The filled ones do not graduate. The rate is 391 in 1,000 among the 3,630 whose outcome is settled, after setting aside 794 still enrolled.

Every one of them costs three things. The student loses the years and the fees. The school loses the tuition. The country loses a graduate it had already half paid for.

The school already knows this is happening. What it does not know is which students. It finds out when the first exam results come back. By then half the year is gone. And the student is usually gone too.

The question is not whether to help. It is who to help first, and when.

2,100 EUR

is the tuition one saved student pays back over a three year degree. A low estimate on purpose. The real loss to the student and the country is larger and harder to pin down. So it is left out.

What the tool does, on the day a student enrolls

It reads the enrollment form. Nothing else.

Age. Course. How they applied. Whether they hold a scholarship. Whether the fees are paid. What their parents do. Everything on that list is known before the student sits in a single class.

It does not look at grades, attendance, or coursework.

That was a deliberate choice and it costs accuracy. A tool that waits for first-term results would be far more certain. It would also be useless. By then the student has already decided.

It hands the tutoring team a ranked list.

Most at risk at the top. Not a label. Not a verdict. Never a number shown to a student. A starting point for a conversation, twelve months early.

The tool does not decide anything. It decides what a person looks at first.

12 months

of warning, instead of none

8 in 10

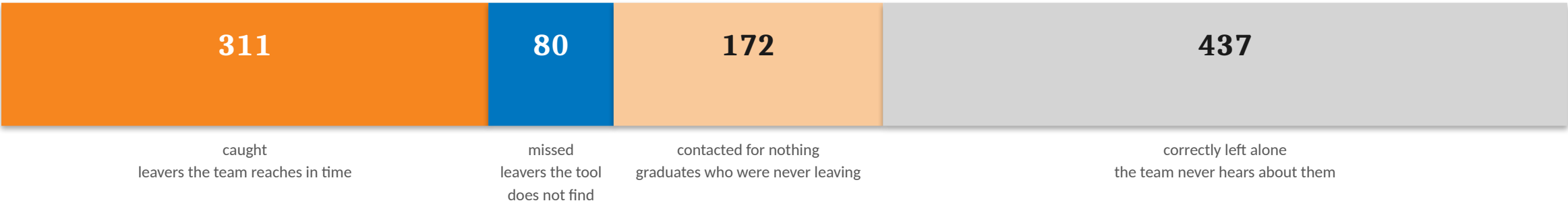
of the students who go on to leave are on the list

1 in 2

students are flagged, sized against what the tutoring team can handle

Out of every 1,000 students

Scaled from 726 real students the tool had never seen. It was tested once, on data held back from the start.



Of the 391 who leave, the tool finds 311 of them.

Eighty are missed. That is the honest number and it will not be zero. No tool that runs on an enrollment form finds everyone. A tool that claimed to would be lying.

The 172 is the number worth arguing about, and slide 6 argues about it.

483

students the tool flags, roughly one in two. That is more than the tutoring team can staff, and the fairness fix on slide 8 is what brings the list back within reach.

Every number on this slide came from students the tool had never seen.

The return

Three assumptions, all stated, all careful. Change any one and the answer moves. That is why the curve is here, not just the number.

A saved student is worth about 2,100 EUR.

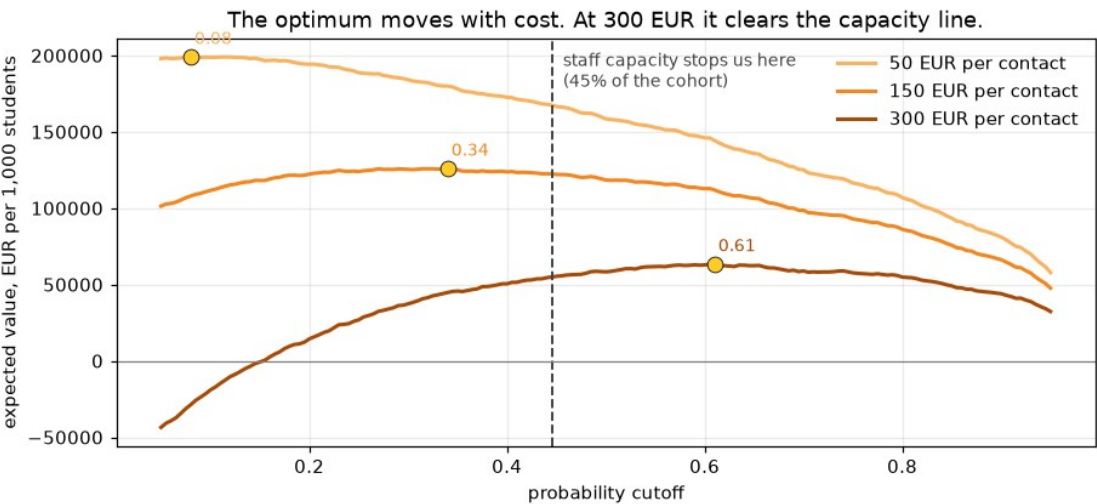
Three years of public tuition. Nothing else counted.

Reaching out works about three times in ten.

So a student the tool finds is worth about 630 EUR on average, not 2,100.

A conversation costs somewhere between 50 and 300 EUR.

The school knows this number. I do not. So all three are shown.



A conversation costs	Best case	At 1 in 2
50 EUR	199,022 EUR	171,942 EUR
150 EUR	125,289 EUR	123,595 EUR
300 EUR	58,967 EUR	51,074 EUR

Value per 1,000 students. Best case is the most profitable cutoff. At 1 in 2 is the cutoff the tool deploys at, and the gap between them at 150 EUR is only 1,700 EUR. So aiming the tool at the team's capacity is nearly free. Slide 8 is the thing that is not free.

At 150 EUR a conversation, a save returns four times what a contact costs. The money says reach out broadly. There is barely any sorting left to do.

The price, and who pays it

172 students in every 1,000 are contacted who were never going to leave.

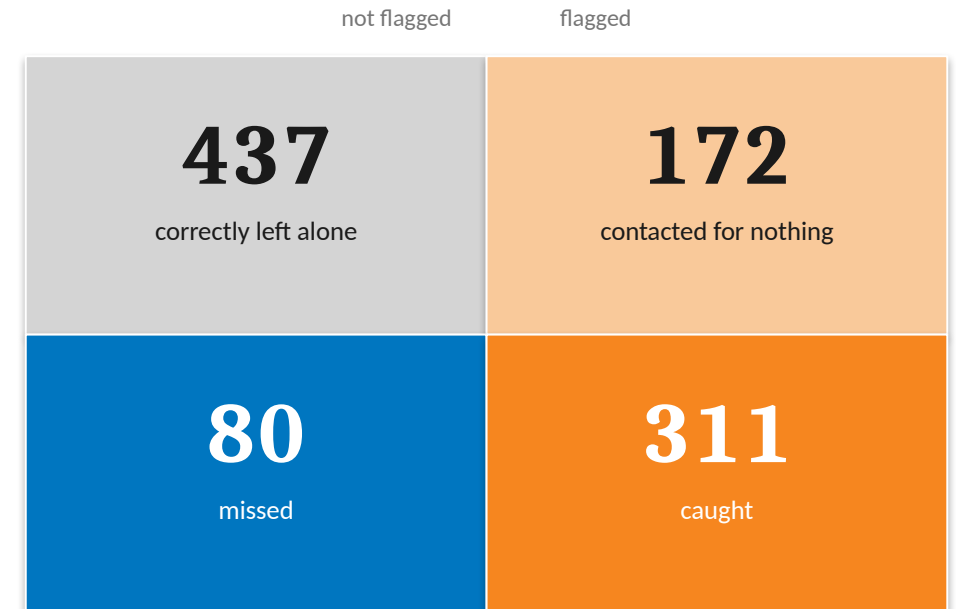
That is the real cost of this tool and it is not paid in euros. It is paid by a student pulled into a help conversation they did not need. A student just told, in effect, that a system thinks they are failing.

The number can be driven down. Flag fewer students and fewer graduates get bothered. But every student cut from the list is also a leaver the team never reaches. And the 80 missed today becomes 120.

This is the trade and it does not go away.

The tool cannot do both at once. Find more leavers and it bothers more graduates. Bother fewer graduates and it finds fewer leavers. Where the school sits is a judgement about students, not a calculation.

Which is why the list goes to a person, and the person decides who to call.



The tool is worse at its job for women

The tool was checked across gender, age, scholarship, and debt. Gender is the one that fails on fairness. The other three flag groups that really do drop out more, though age still needs a closer look.



Share of the students who go on to leave that the tool actually finds.

3 in 10

women who go on to drop out are never on the list at all. For men it is closer to one in eight.

This does not show up in any count of who gets flagged.

The tool flags men twice as often as women. So a check on flagging rates says men are over-contacted and stops there. That check is not wrong. It just answers a different question.

The harm only appears when you stop counting who gets contacted and start counting who gets missed.

The students this tool exists to reach are being missed. And they are being missed unevenly.

Closing that gap costs about 30 students

There is a way to correct it. It works. It is not free, and the school should see the bill before it decides.



Leavers found, per 1,000 students.

Thirty fewer students are reached in every thousand.

That is a real cost and it should be written down plainly, not buried in an appendix. Thirty students who would have been caught now will not be. In money that is about 8,000 EUR per 1,000 students at a 150 EUR contact, and the catch rate falls from about 8 in 10 to 7 in 10.

What it buys is that those thirty are not loaded onto women.

The correction does not make the tool better. It makes it evenly bad.

The list of about 410 fits what the team can reach, so everyone on it is contacted.

The recommendation is to pay it. But the number is on the table, not hidden.

The option the school should not take

A second correction closes the same gap and looks better on paper. Overall accuracy goes up.

It gets there by contacting fewer students of every kind. It closes the gap by helping fewer people. And it costs seventy fewer leavers found instead of thirty.

Fixing a gap by helping fewer people is not a fix.

This is the school's decision, not the tool's.

Thirty students is the price of the tool being as good at finding women as it is at finding men. Nobody but the school can say whether that is worth paying.

How to put it in, without betting the school on it

Nothing here asks anyone to trust the tool on day one. Each phase earns the next.

ONE TERM

1. Watch it, do not act on it

The tool produces its list. Nobody is contacted. At the end of term, compare who it flagged against who actually left. And against who the tutors would have worried about anyway.

If it does not beat the tutors, it does not deploy.

ONE YEAR

2. One department, real contact

A single department runs it for real. The list goes to a tutor. The tutor decides who to call.

This measures the thing nobody has measured yet. Whether reaching out actually keeps anyone. Every number on slide 5 rests on an assumption of three in ten. This is where that gets checked.

ONGOING

3. Full rollout, with a standing check

School-wide, with the gender gap re-measured at every intake and reported to whoever owns this decision.

The gap was measured on one group of 130 women who left. It is not the kind of thing that stays fixed on its own.

The tool is cheap to run and expensive to trust. So buy the trust in stages.

Three conditions, and the ask

1. A student never sees their score.

Not on a portal, not in a letter, not read out by a tutor. A person told they are seventy percent likely to fail has been handed a prediction, not help.

2. Fix what the school can actually change.

The single strongest signals in the tool are which course a student is on and what a parent does, and neither is something a student can change. That is why the tool is a prompt, not a verdict. The strongest signal the school can actually change is whether the fees are paid. A payment plan is cheaper than a tutor and treats the cause, not the symptom.

3. Re-measure the gap every intake.

The gender gap was measured once, on one group. Put it in the annual reporting line and give it an owner.

THE ASK

Approve phase one.

One term. The tool runs, nobody is contacted, and at the end we know whether it beats a tutor's judgement.

It costs a term of nobody's time. And it buys the only thing worth buying at this stage. Evidence.

And approve the shape of the thing.

A ranked list for the tutoring team. Read by a person, acted on by a person. Not a label stuck on a student's record. That is a thing that follows them around.

The tool works. It should not be trusted blindly, and it does not need to be. It only has to point a tutor at the right student, early enough to matter.